# MGM Macau and Cotai

## Resource Efficiency Management Report

August 2022

Energenz Consulting Limited

# EXECUTIVE SUMMARY

Energenz has been working closely with MGM to implement the EMOs as well as identify additional opportunities to achieve ISO certification energy targets. Energenz appreciates the great support and trust from MGM team over the years, in particular the Facility Management and the Sustainability Teams.

## Key Performance Indicators

Energenz reports both normalized and actual savings on a monthly basis for both properties. The normalized savings are determined based on a 2019 baseline, with aligned annual savings targets to meet both ISO50001 as well as Energenz's retainer contract requirements. Please note that the reported normalized savings achieved are to-date and cumulative for each month of the respective year.

**Table 1. ISO and Retainer Contract Energy Performance Targets for 2025**

| MGM Properties | Baseline | Energy Savings | 2021 | Normalized Savings Achieved Up to August 2022 | 2023 | 2024 | 2025 |
|---|---|---|---|---|---|---|---|
| **Macau** | 2019 | Target | -4% | -8% | -12% | N/A | **-13%** |
| | | Actual | **-5.9%** | **-6.1%** | - | - | **-** |
| **Cotai** | 2019 | Target | -4% | -8% | -12% | N/A | **-13%** |
| | | Actual | **-6.8%** | **-10.5%** | - | - | **-** |

**MGM Cotai ISO energy performance: e-bus consumption is included.*

The actual monthly savings are determined against the 2019 baseline consumption and cumulative for each month of 2022.

**Table 2. Internal Actual Energy Performance Targets for 2025**

| MGM Properties | Baseline | Actual Saving Target[1] | Savings Achieved Up to August 2022 |
|---|---|---|---|
| **Macau** | 2019 | 15% | **-9.2%** |
| **Cotai** | 2019 | 15% | **-16.7%** |

[1] Actual Energy Savings Target for 2025 has been established based on future CAPEX projects, following review with MGM teams.

### Highlighted Changes in Operation

**Macau**

1. MGM Macau has resumed normal operation on 23 July 2022, all of the previous low occupancy energy saving measures have been deactivated.
2. Hotel tower L19 PAU dehumidification mode has been switched off on 8 August 2022 to reduce the LPG boiler usage.

**Cotai**

1. MGM Cotai has resumed normal operation on 23 July 2022, all of the previous low occupancy energy saving measures have been deactivated.
2. Installation of 10 units of variable speed drives (VSD) for hotel tower exhaust fans (EF) have been completed by 22 August 2022. The new VSDs have been set at 45 Hz for energy saving and enhancing the positive differential air pressure into the hotel towers staircases.

### Consumption Intensity

The YTD electricity and water use intensities are tracked on a monthly basis for both properties:

**Table 3. Year to Date Actual Electricity Use Intensity (EUI)**

| MGM Properties | YTD EUI (kWh/$m^2$) | % Change Compared to 2021 |
|---|---|---|
| **Macau** | **24.4** | **-3.4%** |
| **Cotai** | **24.2** | **-10.0%** |

**Table 4. Year to Date Actual Water Use Intensity (WUI)**

| MGM Properties | YTD WUI ($m^3$/ $m^2$) | % Change Compared to 2021 |
|---|---|---|
| **Macau** | **0.19** | **-9.7%** |
| **Cotai** | **0.13** | **-23.8%** |

*Note: Cotai Area: 374,138 $m^2$  Macau Area: 211,545 $m^2$*

## ISO 50001 Certification Audit

**Macau**

The HKQAA 2022 surveillance audit was completed on 30 March 2022. There is no nonconformity (NC) and areas for improvement (AFI) from auditors.

**Cotai**

The HKQAA 2022 recertification audit will be scheduled on 29 & 30 September 2022.

## Target Milestones for the Year 2022

The following activities are planned for completion this year:

- To perform ISO 50001 2022 Internal and External Audit for MGM Macau and MGM Cotai
- To deploy and fine-tune the DGLux dashboard with heatmap development extension
- To perform 2021 Carbon and Waste Audit for MGM Macau and MGM Cotai
- To assist MGM with the COTAI phase 2 design and planning
- To investigate and implement the below list of projects during the Year 2022

**Table 5. MGM Macau Year 2022 Projects**

| Item | Initiatives |
|---|---|
| **Electricity & LPG** | |
| 1 | Chilled Water Pump Retrofit (Phase 1) |
| 2 | AHU Static Pressure Set-Point Reset (Phase 2) |
| 3 | PICCV Implementation (Phase 3) |
| 4 | Air Side Equipment EC Fan/Motor Replacement |
| 5 | VAV Space Temperature Setpoint Optimization with Heatmap Control |
| 6 | Heatmap Phasing Implementation (Stage 3, 9a, 9b, 10a) |
| 7 | Neon Light LED Replacement (Phase 2) |
| 8 | Replacement of L2M Steam-to-Water Heat Exchanger to Fully Utilize Heat Pump |
| **Water** | |
| 9 | CT Conductivity Blowdown Optimization |

**Table 6. MGM Cotai Year 2022 Projects**

| Item | Initiatives |
|---|---|
| **Electricity & NG** | |
| 1 | Chiller Plant Control Optimization (CHWP/CDWP/CT) |
| 2 | Hot Water System Upgrade<br>1. Increase Max. No of Plate for Pre-heat Compact Unit<br>2. Upgrade Booster Compact Unit<br>3. Install Electric Heater for Booster Calorifier |
| 3 | AHU Static Pressure & Supply Air Temperature Set-Point Reset (Phase 2) |
| 4 | PICCV Implementation (Phase 3) |
| 5 | Hotel Towers Building Pressurization and EAF Optimization |
| 6 | Kitchen Exhaust Fan (KEF) Steps Speed Control Optimization |
| 7 | Carpark Ventilation Fans Control Optimization with IoT Sensor Installation (Phase 2) |
| 8 | VAV Space Temperature Setpoint Optimization with Heatmap Control |
| 9 | Heatmap Phasing Implementation (Stage 5, 6, 7, 8a, 8b, 10b) |
| 10 | Centralized RO Pump Set Replacement (10 Units) |
| **Water** | |
| 11 | Extension of CDP Water Recycling |
| 12 | Rainwater Harvesting for CT Make Up (Phase 1) |
| 13 | Non-potable Centralized RO Optimization for Cooling Tower |

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# ACKNOWLEDGEMENT

Energenz has been working with MGM Team as the energy manager to achieve energy savings in MGM properties and feed operational intelligence into the design.

The major goal of this project is not only to realize operation energy savings but also to extend the life of equipment, hence reducing CAPEX. MGM has demonstrated continuous commitment to be an environmentally and socially responsible business in the past years. In addition to that Energenz has assisted MGM Macau and Cotai to achieve several other energy management and sustainability awards and certifications, such as ISO 50001, CEM Energy Savings Award, and China 3 Star Green Building Operation Label.

Energenz appreciates the great support and trust from MGM Team over the years, especially the Facilities Management and the Sustainability Teams.

# 1. Portfolio Performance Summary

## 1.1 Electricity Use Intensity (EUI)

Macau's electricity use intensity (EUI) in August 2022 had decreased by **5.8%** compared to the same month in 2021 and decreased by **3.4%** in 2022 compared to 2021. Meanwhile, Cotai's EUI in August 2022 has decreased by **13.7%** compared to the same month in 2021 and decreased by **10.0%** in 2022 compared to 2021.

**Table 7. Actual Electricity Consumption Intensity (EUI) Comparison**

| Monthly Period | Macau | | | Cotai | | |
|---|---|---|---|---|---|---|
| | Year 2021 Actual EUI (kWh/m²) | Year 2022 Actual EUI (kWh/m²) | % Change | Year 2021 Actual EUI (kWh/m²) | Year 2022 Actual EUI (kWh/m²) | % Change |
| **January** | 20.4 | 22.6 | +11.0% | 21.8 | 22.3 | +2.1% |
| **February** | 19.8 | 20.2 | +1.8% | 20.4 | 20.1 | -1.4% |
| **March** | 23.3 | 24.4 | +4.5% | 24.6 | 23.6 | -4.0% |
| **April** | 23.8 | 24.3 | +1.8% | 25.4 | 24.0 | -5.6% |
| **May** | 28.9 | 25.9 | -10.5% | 30.6 | 25.6 | -16.3% |
| **June** | 28.7 | 27.3 | -4.9% | 29.7 | 27.0 | -9.3% |
| **July** | 30.8 | 25.9 | -15.8% | 31.7 | 24.3 | -23.4% |
| **August** | 29.8 | 28.1 | -5.8% | 30.5 | 26.3 | -13.7% |
| **September** | | | | | | |
| **October** | | | | | | |
| **November** | | | | | | |
| **December** | | | | | | |
| **Average** | **25.7** | **24.8** | **-3.4%** | **26.8** | **24.2** | **-10.0%** |

## 1.2 Water Use Intensity (WUI)

Macau's water use intensity (WUI) in August 2022 decreased by **3.8%** compared to the same month in 2021 and decreased by **9.7%** in 2022 compared to 2021. Meanwhile, Cotai's WUI in August 2022 decreased by **21.8%** compared to the same month in 2021 and decreased by **23.8%** for 2022 compared to 2021.

**Table 8. Actual Water Consumption Intensity (WUI) Comparison**

| Monthly Period | Macau | | | Cotai | | |
|---|---|---|---|---|---|---|
| | Year 2021 Actual WUI ($m^3/m^2$) | Year 2022 Actual WUI ($m^3/m^2$) | % Change | Year 2021 Actual WUI ($m^3/m^2$) | Year 2022 Actual WUI ($m^3/m^2$) | % Change |
| **January** | 0.19 | 0.18 | -7.2% | 0.116 | 0.119 | +3.2% |
| **February** | 0.19 | 0.20 | +1.1% | 0.14 | 0.12 | -10.9% |
| **March** | 0.20 | 0.19 | -7.4% | 0.15 | 0.12 | -19.3% |
| **April** | 0.24 | 0.21 | -9.1% | 0.18 | 0.14 | -21.2% |
| **May** | 0.25 | 0.23 | -7.7% | 0.21 | 0.15 | -28.2% |
| **June** | 0.21 | 0.19 | -9.3% | 0.18 | 0.12 | -30.7% |
| **July** | 0.22 | 0.15 | -32.6% | 0.19 | 0.10 | -46.2% |
| **August** | 0.20 | 0.19 | -3.8% | 0.17 | 0.13 | -21.8% |
| **September** | | | | | | |
| **October** | | | | | | |
| **November** | | | | | | |
| **December** | | | | | | |
| **Average** | **0.21** | **0.19** | **-9.7%** | **0.16** | **0.13** | **-23.8%** |

# 2. MGM Macau Performance

## 2.1 Electricity Consumption

The actual electricity consumption and cost in August 2022 decreased by **5.8%** and **2.0%** respectively compared to the same month in 2021. To date, MGM Macau has decreased the actual electricity consumption by **3.4%** and associated cost increased by **0.6%** respectively.

**Table 9. Macau - Summary of Actual Monthly Electricity Consumption and Cost**

| Monthly Period | Actual Electricity Consumption (kWh) | | | Actual Electricity Cost (MOP) | | | % Change – CDD |
|---|---|---|---|---|---|---|---|
| | 2021 | 2022 | % Change - Actual | 2021 | 2022 | % Change - Actual Cost | |
| **January** | 4,310,160 | 4,784,280 | +11.0% | 4,717,260 | 5,519,350 | +17.0% | +15.8% |
| **February** | 4,190,760 | 4,266,240 | +1.8% | 4,591,170 | 4,942,590 | +7.7% | -90.8% |
| **March** | 4,931,880 | 5,152,320 | +4.5% | 5,387,480 | 5,938,160 | +10.2% | -6.9% |
| **April** | 5,042,160 | 5,131,680 | +1.8% | 5,808,340 | 6,183,420 | +6.5% | -11.4% |
| **May** | 6,122,640 | 5,478,840 | -10.5% | 7,034,530 | 6,595,030 | -6.2% | -32.6% |
| **June** | 6,072,960 | 5,777,040 | -4.9% | 7,681,070 | 7,567,480 | -1.5% | -1.0% |
| **July** | 6,506,760 | 5,480,880 | -15.8% | 8,217,570 | 7,172,780 | -12.7% | +4.1% |
| **August** | 6,306,840 | 5,938,920 | -5.8% | 7,960,170 | 7,799,020 | -2.0% | +1.7% |
| **September** | | | | | | | |
| **October** | | | | | | | |
| **November** | | | | | | | |
| **December** | | | | | | | |
| **Total/Average** | **43,484,160** | **42,010,200** | **-3.4%** | **51,397,590** | **51,717,830** | **+0.6%** | **-10.3%** |

***Important note:*** *Cooling Degree Days (CDDs) is the main variable taken into consideration on Electricity consumption normalization.*

Energenz summarized the key items which could result in electricity consumption decrease below:

- Chiller Plant electricity consumption has decreased by 3.5% in August 2022 compared to the same month in 2021 despite the 1.7% increase in CDD.

Table 10 presents the comparison of electricity rates in the years 2021 and 2022. The electricity rate in August 2022 increased by **4.0%** compared to the same month in 2021.

**Table 10. Macau - Summary of Actual Monthly Electricity Rate**

| Monthly Period | Year 2021 – Electricity Rate (MOP/kWh) | Year 2022 – Electricity Rate (MOP/kWh) | % Change – Electricity Rate |
|---|---|---|---|
| **January** | 1.09 | 1.15 | +5.4% |
| **February** | 1.10 | 1.16 | +5.7% |
| **March** | 1.09 | 1.15 | +5.5% |
| **April** | 1.15 | 1.20 | +4.6% |
| **May** | 1.15 | 1.20 | +4.8% |
| **June** | 1.26 | 1.31 | +3.6% |
| **July** | 1.26 | 1.31 | +3.6% |
| **August** | 1.26 | 1.31 | +4.0% |
| **September** | | | |
| **October** | | | |
| **November** | | | |
| **December** | | | |
| **Total/Average** | **1.17** | **1.23** | **+4.6%** |

Table 11 summarizes the monthly normalized electricity consumption. The actual electricity consumption in August 2022 decreased by **10.4%** compared to 2019 normalized consumption in the same month.

**Table 11. Macau - Summary of Normalized Monthly Electricity Consumption**

| Monthly Period | 2019 Normalized Electricity Consumption (kWh) | 2022 Actual Electricity Consumption (kWh) | % Change - Normalized |
|---|---|---|---|
| **January** | 4,790,450 | 4,784,280 | -0.1% |
| **February** | 4,631,715 | 4,266,240 | -7.9% |
| **March** | 5,342,712 | 5,152,320 | -3.6% |
| **April** | 5,631,086 | 5,131,680 | -8.9% |
| **May** | 5,930,696 | 5,478,840 | -7.6% |
| **June** | 6,468,840 | 5,777,040 | -10.7% |
| **July** | 6,859,773 | 5,480,880 | -20.1% |
| **August** | 6,628,440 | 5,938,920 | -10.4% |
| **September** | | | |
| **October** | | | |
| **November** | | | |
| **December** | | | |
| **Total/Average** | **46,283,713** | **42,010,200** | **-9.2%** |

## 2.2 LPG Consumption

LPG consumption can be broken down into three (3) categories: boiler, hotel upper zone DHW, and kitchens/restaurants. Since the boiler and restaurants LPG utilization cover over 95% of the overall site LPG consumption, Energenz focuses the analysis on those two areas.

According to Table 12, the actual LPG consumption and cost in August 2022 decreased by **8.8%** and **0.1%** respectively compared to the same month in 2021. To date, MGM Macau has increased the actual LPG consumption by **12.6%** and associated costs increased by **45.0%**.

**Table 12. Macau - Summary of Actual Monthly LPG Consumption and Cost**

| Monthly Period | Actual LPG Consumption (kg) | | | Actual LPG Cost (MOP) | | | % Change – Min OAT |
|---|---|---|---|---|---|---|---|
| | 2021 | 2022 | % Change - Actual | 2021 | 2022 | % Change - Actual Cost | |
| **January** | 121,398 | 121,221 | -0.1% | 947,708 | 1,327,490 | +40.1% | +56.0% |
| **February** | 74,797 | 146,623 | +96.0% | 632,778 | 1,450,103 | +129.2% | -52.6% |
| **March** | 84,131 | 91,040 | +8.2% | 711,751 | 900,382 | +26.5% | -5.4% |
| **April** | 77,702 | 85,743 | +10.3% | 652,357 | 858,756 | +31.6% | -28.6% |
| **May** | 71,294 | 82,704 | +16.0% | 578,906 | 992,447 | +71.4% | -26.8% |
| **June** | 62,104 | 77,044 | +24.1% | 504,285 | 924,531 | +83.3% | -1.2% |
| **July** | 71,398 | 44,991 | -37.0% | 579,753 | 539,887 | -6.9% | -0.8% |
| **August** | 74,204 | 67,705 | -8.8% | 698,258 | 697,361 | -0.1% | +0.4% |
| **September** | | | | | | | |
| **October** | | | | | | | |
| **November** | | | | | | | |
| **December** | | | | | | | |
| **Total/Average** | **637,028** | **717,070** | **+12.6%** | **5,305,797** | **7,690,956** | **+45.0%** | **-10.7%** |

***Important note:*** *Minimum Outdoor Air Temperature (OAT) is the main variable taken into consideration on LPG consumption normalization.*

Energenz summarized the key items which could result in LPG consumption decrease below:

- Kitchen LPG consumption decreased by 12.1% and food covers decreased by 17.7% in August 2022 compared with the same month in 2021.
- L36 Water Heater LPG Consumption has decreased by 9.0% in August 2022 compared with the same month in 2021.

Table 13 presents the comparison of LPG rates in the years 2021 and 2022. The LPG rate in August 2022 increased by **9.5%** compared with the same month in 2021.

**Table 13. Macau - Summary of Actual Monthly LPG Rate**

| Monthly Period | Year 2021 – LPG Rate (MOP/kg) | Year 2022 – LPG Rate (MOP/ kg) | % Change – LPG Rate |
|---|---|---|---|
| **January** | 7.81 | 10.95 | +40.3% |
| **February** | 8.46 | 9.89 | +16.9% |
| **March** | 8.46 | 9.89 | +16.9% |
| **April** | 8.40 | 10.02 | +19.3% |
| **May** | 8.12 | 12.00 | +47.8% |
| **June** | 8.12 | 12.00 | +47.8% |
| **July** | 8.12 | 12.00 | +47.8% |
| **August** | 9.41 | 10.30 | +9.5% |
| **September** | | | |
| **October** | | | |
| **November** | | | |
| **December** | | | |
| **Total/Average** | **8.36** | **10.88** | **+30.1%** |

Table 14 summarizes the monthly normalized LPG consumption. The actual LPG consumption in August 2022 increased by **4.3%** compared to 2019 normalized consumption in the same month.

**Table 14. Macau - Summary of Normalized Monthly LPG Consumption**

| Monthly Period | 2019 Normalized LPG Consumption (kg) | 2022 Actual LPG Consumption (kg) | % Change – Normalized |
|---|---|---|---|
| **January** | 95,134 | 121,221 | +27.4% |
| **February** | 117,728 | 146,623 | +24.5% |
| **March** | 78,544 | 91,040 | +15.9% |
| **April** | 85,634 | 85,743 | +0.1% |
| **May** | 70,235 | 82,704 | +17.8% |
| **June** | 72,880 | 77,044 | +5.7% |
| **July** | 66,446 | 44,991 | -32.3% |
| **August** | 64,919 | 67,705 | +4.3% |
| **September** | | | |
| **October** | | | |
| **November** | | | |
| **December** | | | |
| **Total/Average** | **651,521** | **717,070** | **+10.1%** |

## 2.3 ISO50001 Energy Consumption Record

The Macau ISO50001 energy performance target is to achieve 13% normalized energy savings by 2025 compared to the 2019 baseline. To date, normalized energy saving achieved up to August 2022 is **6.1%** as shown in Table 15. Table 16 below summarizes the site monthly energy consumption (Electricity and LPG combined) compared to the 2019 baseline.

**Table 15. Macau ISO 50001 Normalized Energy Performance Targets for 2025**

| MGM Properties | Baseline | Energy Savings | 2021 | Normalized Savings Achieved Up to August 2022 | 2023 | 2024 | 2025 |
|---|---|---|---|---|---|---|---|
| **Macau** | 2019 | Target | -4% | -8% | -12% | N/A | **-13%** |
| | | Actual | **-5.9%** | **-6.1%** | - | - | - |

**Table 16. Macau ISO 50001 Normalized Monthly Energy Consumption**

| Monthly Period | 2019 Normalized Energy Consumption (GJ) | 2022 Actual Energy Consumption (GJ) | % Change – Normalized vs Actual |
|---|---|---|---|
| **January** | 21,555 | 22,715 | +5.4% |
| **February** | 22,007 | 22,000 | -0.03% |
| **March** | 22,792 | 22,672 | -0.5% |
| **April** | 24,151 | 22,358 | -7.4% |
| **May** | 24,532 | 23,470 | -4.3% |
| **June** | 26,589 | 24,287 | -8.7% |
| **July** | 27,705 | 21,769 | -21.4% |
| **August** | 26,803 | 24,447 | -8.8% |
| **September** | | | |
| **October** | | | |
| **November** | | | |
| **December** | | | |
| **Total/Average** | **196,135** | **183,720** | **-6.3%** |

***Important note:*** *Factors used for fuels' unit conversion: 1) Electricity (0.0036) and LPG (0.0453)*

## 2.4 Water Consumption

Based on the existing water metering system, the overall site water consumption can be broken down into four (4) categories: Kitchen and Restaurants; Hotel/Tower; Fire Services; and Chiller Plant & miscellaneous[2].

According to Table 17, the actual water consumption and cost in August 2022 both decreased by **3.8%** compared to the same month in 2021. To date, MGM Macau has decreased actual water consumption and associated costs by **9.7%** and **9.5%** respectively.

**Table 17. Macau - Summary of Actual Monthly Water Consumption and Cost**

| Monthly Period | Actual Water Consumption ($m^3$) | | | Actual Water Cost (MOP) | | | Normalization Factors | |
|---|---|---|---|---|---|---|---|---|
| | 2021 | 2022 | % Change Actual | 2021 | 2022 | % Change Actual Cost | % Change – CDD | % Change – Food Covers |
| **January** | 39,952 | 37,078 | -7.2% | 299,330 | 287,500 | -4.0% | +15.8% | +36.7% |
| **February** | 41,079 | 41,524 | +1.1% | 317,780 | 322,270 | +1.4% | -90.8% | +43.2% |
| **March** | 43,168 | 39,962 | -7.4% | 334,000 | 310,160 | -7.1% | -6.9% | -9.6% |
| **April** | 49,819 | 45,295 | -9.1% | 386,600 | 351,540 | -9.1% | -11.4% | -11.4% |
| **May** | 53,145 | 49,042 | -7.7% | 420,070 | 380,590 | -9.4% | -32.6% | -12.2% |
| **June** | 44,508 | 40,383 | -9.3% | 345,320 | 313,390 | -9.2% | -1.0% | -35.0% |
| **July** | 46,227 | 31,179 | -32.6% | 358,640 | 242,100 | -32.5% | +4.1% | -92.5% |
| **August** | 42,681 | 41,073 | -3.8% | 331,220 | 318,730 | -3.8% | +1.7% | -17.7% |
| **September** | | | | | | | | |
| **October** | | | | | | | | |
| **November** | | | | | | | | |
| **December** | | | | | | | | |
| **Total/Average** | **360,579** | **325,536** | **-9.7%** | **2,792,960** | **2,526,280** | **-9.5%** | **-10.3%** | **-15.7%** |

***Important note:*** *Cooling Degree Days (CDDs) and Food Covers are the main variables taken into consideration on Water consumption normalization.*

Energenz summarized the key items which could result in water consumption decrease below:

- Podium water consumption for August 2022 decreased by 10.3% compared to the same month in 2021.

[2] Miscellaneous includes the water consumption from toilets, laundry, etc.

Table 18 presents the comparison of water rates in the years 2021 and 2022. The water rate in August 2022 remains unchanged compared to the same month in 2021. The average utility rate of water has increased by **0.3%** compared to 2021.

**Table 18. Macau - Summary of Actual Monthly Water Rates**

| Monthly Period | Year 2021 – Water Rate (MOP/m³) | Year 2022 – Water Rate (MOP/m³) | % Change – Water Rate |
|---|---|---|---|
| **January** | 7.49 | 7.75 | +3.5% |
| **February** | 7.74 | 7.76 | +0.3% |
| **March** | 7.74 | 7.76 | +0.3% |
| **April** | 7.760 | 7.761 | +0.01% |
| **May** | 7.90 | 7.76 | -1.8% |
| **June** | 7.759 | 7.76 | +0.02% |
| **July** | 7.758 | 7.765 | +0.09% |
| **August** | 7.76 | 7.76 | 0% |
| **September** | | | |
| **October** | | | |
| **November** | | | |
| **December** | | | |
| **Total/Average** | **7.74** | **7.76** | **+0.3%** |

Table 19 summarized the monthly normalized water consumption. The actual water consumption in August 2022 decreased by **5.1%** compared to 2019 normalized consumption in the same month.

**Table 19. Macau - Summary of Normalized Monthly Water Consumption**

| Monthly Period | 2019 Normalized Water Consumption (m³) | 2022 Actual Water Consumption (m³) | % Change - Normalized |
|---|---|---|---|
| **January** | 35,691 | 37,078 | +3.9% |
| **February** | 37,518 | 41,524 | +10.7% |
| **March** | 37,955 | 39,962 | +5.3% |
| **April** | 39,506 | 45,295 | +14.7% |
| **May** | 42,691 | 49,042 | +14.9% |
| **June** | 41,208 | 40,383 | -2.0% |
| **July** | 36,632 | 31,179 | -14.9% |
| **August** | 43,294 | 41,073 | -5.1% |
| **September** | | | |
| **October** | | | |
| **November** | | | |
| **December** | | | |
| **Total/Average** | **314,495** | **325,536** | **+3.5%** |

## 2.5 Macau Year 2022 Projects

Energenz will continue to perform analysis and coordinate with MGM to implement the following projects in 2022 and reduce the site energy and water consumption which supports Energenz annual normalized energy savings target.

**Table 20. MGM Macau Year 2022 Projects**

| Item | Initiatives |
|---|---|
| **Electricity & LPG** | |
| 1 | Chilled Water Pump Retrofit (Phase 1) |
| 2 | AHU Static Pressure Set-Point Reset (Phase 2) |
| 3 | PICCV Implementation (Phase 3) |
| 4 | Air Side Equipment EC Fan/Motor Replacement |
| 5 | VAV Space Temperature Setpoint Optimization with Heatmap Control |
| 6 | Heatmap Phasing Implementation (Stage 3, 9a, 9b, 10a) |
| 7 | Neon Light LED Replacement (Phase 2) |
| 8 | Replacement of L2M Steam-to-Water Heat Exchanger to Fully Utilize Heat Pump |
| **Water** | |
| 9 | CT Conductivity Blowdown Optimization |

# 3. MGM Cotai Performance

## 3.1 Electricity Consumption

The actual electricity consumption in August 2022 decreased by **13.7%** and the cost decreased by **9.9%** compared to the same month in 2021. To date, MGM Cotai has decreased actual electricity consumption by **10%** and associated costs decreased by **6.0%**.

**Table 21. Cotai - Summary of Actual Monthly Electricity Consumption and Cost**

| Monthly Period | Actual Electricity Consumption (kWh) | | | Actual Electricity Cost (MOP) | | | % Change – CDD |
|---|---|---|---|---|---|---|---|
| | 2021 | 2022 | % Change - Actual | 2021 | 2022 | % Change - Actual Cost | |
| **January** | 8,168,400 | 8,336,400 | +2.1% | 9,315,910 | 10,013,460 | +7.5% | +15.8% |
| **February** | 7,637,640 | 7,528,560 | -1.4% | 8,737,740 | 9,066,060 | +3.8% | -90.8% |
| **March** | 9,197,160 | 8,833,200 | -4.0% | 10,499,490 | 10,632,040 | +1.3% | -6.9% |
| **April** | 9,517,560 | 8,985,360 | -5.6% | 11,430,660 | 11,262,370 | -1.5% | -11.4% |
| **May** | 11,446,440 | 9,585,000 | -16.3% | 13,735,880 | 11,983,070 | -12.8% | -32.6% |
| **June** | 11,122,320 | 10,087,080 | -9.3% | 13,340,650 | 12,615,080 | -5.4% | -1.0% |
| **July** | 11,853,120 | 9,082,680 | -23.4% | 14,194,900 | 11,346,140 | -20.1% | +4.1% |
| **August** | 11,412,960 | 9,847,320 | -13.7% | 13,675,130 | 12,318,540 | -9.9% | +1.7% |
| **September** | | | | | | | |
| **October** | | | | | | | |
| **November** | | | | | | | |
| **December** | | | | | | | |
| **Total/Average** | **80,355,600** | **72,285,600** | **-10.0%** | **94,930,360** | **89,236,760** | **-6.0%** | **-10.3%** |

***Important note**: Cooling Degree Days (CDDs) is the main variable taken into consideration on Electricity consumption normalization. The e-Bus electricity consumption has been included with the presented electricity consumption above.*

Energenz summarized the key items which could result in electricity consumption decrease below:

- Chiller plant consumption has decreased by 20.8% compared to the same month in 2021 due to the changes in chiller plant operations during the COVID low occupancy periods.
- Airside equipment consumption has decreased by 8.9% compared to the same month in 2021 due to the changes in the operation of the airside equipment during the COVID low occupancy periods.
- Hotel Tower consumption has decreased by 38.9% compared to the same month in 2021 due to the COVID low occupancy.

Table 22 presents the monthly e-bus electricity consumption in MGM Cotai. The e-bus electricity consumption in August 2022 decreased by **29.5%** compared to the same month in 2021

**Table 22. Cotai - Summary of Monthly e-bus Electricity Consumption**

| Monthly Period | E-bus Electricity Consumption (kWh) | | |
|---|---|---|---|
| | 2021 | 2022 | % Change - Actual |
| **January** | 44,839 | 42,287 | -5.7% |
| **February** | 32,534 | 47,905 | +47.2% |
| **March** | 34,902 | 38,793 | +11.1% |
| **April** | 37,412 | 38,537 | +3.0% |
| **May** | 51,584 | 51,328 | -0.5% |
| **June** | 56,629 | 36,420 | -35.7% |
| **July** | 64,410 | 0 | -100.0% |
| **August** | 44,185 | 31,160 | -29.5% |
| **September** | | | |
| **October** | | | |
| **November** | | | |
| **December** | | | |
| **Total/Average** | **366,494** | **286,428** | **-21.8%** |

Table 23 presents the comparison of electricity rates in the years 2021 and 2022. The electricity rate in August 2022 increased by **4.4%** compared to the same month in 2021.

**Table 23. Cotai - Summary of Actual Monthly Electricity Rates**

| Monthly Period | Year 2021 – Electricity Rate (MOP/kWh) | Year 2022 – Electricity Rate (MOP/kWh) | % Change – Electricity Rate |
|---|---|---|---|
| **January** | 1.14 | 1.20 | +5.3% |
| **February** | 1.14 | 1.20 | +5.3% |
| **March** | 1.14 | 1.20 | +5.4% |
| **April** | 1.20 | 1.25 | +4.4% |
| **May** | 1.20 | 1.25 | +4.2% |
| **June** | 1.20 | 1.25 | +4.3% |
| **July** | 1.20 | 1.25 | +4.3% |
| **August** | 1.20 | 1.25 | +4.4% |
| **September** | | | |
| **October** | | | |
| **November** | | | |
| **December** | | | |
| **Total/Average** | **1.18** | **1.23** | **+4.7%** |

Table 24 summarized the monthly normalized electricity consumption. The actual electricity consumption in August 2022 decreased by **13.8%** compared to 2019 normalized consumption in the same month.

**Table 24. Cotai - Summary of Normalized Monthly Electricity Consumption**

| Monthly Period | 2019 Normalized Electricity Consumption (kWh) | 2022 Actual Electricity Consumption (kWh) | % Change - Normalized |
|---|---|---|---|
| **January** | 8,635,397 | 8,336,400 | -3.5% |
| **February** | 8,394,419 | 7,528,560 | -10.3% |
| **March** | 9,473,790 | 8,833,200 | -6.8% |
| **April** | 9,911,574 | 8,985,360 | -9.3% |
| **May** | 10,366,415 | 9,585,000 | -7.5% |
| **June** | 11,183,378 | 10,087,080 | -9.8% |
| **July** | 11,776,857 | 9,082,680 | -22.9% |
| **August** | 11,425,668 | 9,847,320 | -13.8% |
| **September** | | | |
| **October** | | | |
| **November** | | | |
| **December** | | | |
| **Total/Average** | **81,167,498** | **72,285,600** | **-10.9%** |

## 3.2 Natural Gas Consumption

The actual natural gas consumption and cost in August 2022 decreased by **27.5%** and **23.6%** respectively compared to the same month in 2021. To date, MGM Cotai has decreased actual natural gas consumption and associated costs by **40.9%** and **37.4%** respectively.

**Table 25. Cotai - Summary of Actual Monthly Natural Gas Consumption and Cost**

| Monthly Period | Actual Natural Gas Consumption ($m^3$) | | | Actual Natural Gas Cost (MOP) | | | % Change – Min OAT | % Change – Food Covers |
|---|---|---|---|---|---|---|---|---|
| | 2021 | 2022 | % Change - Actual | 2021 | 2022 | % Change - Actual Cost | | |
| **January** | 60,643 | 59,378 | -2.1% | 363,990 | 376,240 | +3.4% | +56.0% | +22.3% |
| **February** | 54,221 | 76,089 | +40.3% | 325,930 | 482,130 | +47.9% | -52.6% | +31.6% |
| **March** | 68,066 | 52,146 | -23.4% | 407,980 | 330,410 | -19.0% | -5.4% | -34.3% |
| **April** | 86,866 | 43,900 | -49.5% | 518,960 | 278,170 | -46.4% | -28.6% | -37.5% |
| **May** | 90,102 | 45,632 | -49.4% | 537,940 | 289,140 | -46.3% | -26.8% | -48.1% |
| **June** | 80,606 | 33,949 | -57.9% | 482,270 | 215,120 | -55.4% | -1.2% | -55.1% |
| **July** | 102,473 | 3,041 | -97.0% | 610,450 | 19,270 | -96.8% | -0.8% | -92.5% |
| **August** | 51,837 | 37,574 | -27.5% | 311,790 | 238,080 | -23.6% | +0.4% | -34.6% |
| **September** | | | | | | | | |
| **October** | | | | | | | | |
| **November** | | | | | | | | |
| **December** | | | | | | | | |
| **Total/Average** | **594,814** | **351,709** | **-40.9%** | **3,559,310** | **2,228,560** | **-37.4%** | **-10.7%** | **-35.4%** |

***Important note:*** *Minimum Outdoor Air Temperature (OAT is the main variable taken into consideration on natural gas consumption normalization.*

Energenz summarized the key items which could result in natural gas consumption decrease below:

- Kitchen NG consumption has decreased by 32.9% and food cover has decreased by 34.6% compared to the same month in 2021.
- NG boiler consumption has decreased by 99.92% compared to the same month in 2021.

Table 26 presents the comparison of natural gas rates in the years 2021 and 2022. The natural gas rate in August 2022 increased by **5.3%** compared with the same month in 2021.

**Table 26. Cotai - Summary of Actual Monthly Natural Gas Rates**

| Monthly Period | Year 2021 – Natural Gas Rate (MOP/m³) | Year 2022 – Natural Gas Rate (MOP/m³) | % Change – Natural Gas Rate |
|---|---|---|---|
| **January** | 6.00 | 6.34 | +5.6% |
| **February** | 6.01 | 6.34 | +5.4% |
| **March** | 5.99 | 6.34 | +5.7% |
| **April** | 5.97 | 6.34 | +6.1% |
| **May** | 5.97 | 6.34 | +6.1% |
| **June** | 5.98 | 6.34 | +5.9% |
| **July** | 5.96 | 6.34 | +6.4% |
| **August** | 6.01 | 6.34 | +5.3% |
| **September** | | | |
| **October** | | | |
| **November** | | | |
| **December** | | | |
| **Total/Average** | **5.99** | **6.34** | **+5.8%** |

Table 27 summarized the monthly normalized natural gas consumption. The actual natural gas consumption in August 2022 decreased by **63.8%** compared to 2019 normalized consumption in the same month. The normalized model and comparison of natural gas has been modified and included since January 2022.

**Table 27. Cotai - Summary of Normalized Monthly Natural Gas Consumption**

| Monthly Period | 2019 Normalized Natural Gas Consumption (m³) | 2022 Actual Natural Gas Consumption (m³) | % Change - Normalized |
|---|---|---|---|
| **January** | 125,664 | 59,378 | -52.7% |
| **February** | 132,723 | 76,089 | -42.7% |
| **March** | 119,389 | 52,146 | -56.3% |
| **April** | 122,527 | 43,900 | -64.2% |
| **May** | 116,565 | 45,632 | -60.9% |
| **June** | 105,271 | 33,949 | -67.8% |
| **July** | 103,702 | 3,041 | -97.1% |
| **August** | 103,859 | 37,574 | -63.8% |
| **September** | | | |
| **October** | | | |
| **November** | | | |
| **December** | | | |
| **Total/Average** | **929,700** | **351,709** | **-62.2%** |

## 3.3 ISO50001 Energy Consumption Record

Cotai ISO50001 energy target is to achieve 13% normalized energy consumption reduction by 2025 compared to the 2019 baseline. Based on the official agreement with MGM Cotai, e-bus consumption should be included in the energy consumption tracking as part of the ISO50001 requirement. To date, normalized energy saving achieved up to August 2022 is **10.5%** as shown in Table 28 and Table 29 below summarizes the site monthly energy consumption (Electricity and Natural Gas combined) compared to the 2019 baseline.

**Table 28. Cotai ISO 50001 Normalized Energy Performance Targets for 2025**

| MGM Properties | Baseline | Energy Savings | 2021 | Normalized Savings Achieved Up to August 2022 | 2023 | 2024 | 2025 |
|---|---|---|---|---|---|---|---|
| **Cotai** | 2019 | Target | -4% | -8% | -12% | N/A | **-13%** |
| | | Actual | **-6.8%** | **-10.5%** | - | - | - |

**Table 29. Cotai ISO 50001 Normalized Monthly Energy Consumption**

| Monthly Period | 2019 Normalized Energy Consumption (GJ) | 2022 Actual Energy Consumption (GJ) | % Change - Normalized vs Actual |
|---|---|---|---|
| **January** | 35,448 | 32,071 | -9.5% |
| **February** | 34,825 | 29,743 | -14.6% |
| **March** | 38,248 | 33,609 | -12.1% |
| **April** | 39,933 | 33,871 | -15.2% |
| **May** | 41,364 | 36,089 | -12.8% |
| **June** | 43,913 | 37,492 | -14.6% |
| **July** | 45,995 | 32,803 | -28.7% |
| **August** | 44,736 | 36,754 | -17.8% |
| **September** | | | |
| **October** | | | |
| **November** | | | |
| **December** | | | |
| **Total/Average** | **324,464** | **272,432** | **-16.0%** |

***Important note**: Factors used for fuels' unit conversion: 1) Electricity (0.0036) and Natural Gas (0.0347), e-bus consumption is included*

## 3.4 Water Consumption

According to Table 30, the actual water consumption and associated costs in August 2022 decreased by **21.8%** and **21.7%** respectively compared to the same month in 2021. To date, MGM Cotai has decreased actual water consumption and associated costs by **23.8%** and **24%** respectively.

**Table 30. Cotai - Summary of Actual Monthly Water Consumption and Cost**

| Monthly Period | Actual Water Consumption ($m^3$) | | | Actual Water Cost (MOP) | | | Normalization Factors | |
|---|---|---|---|---|---|---|---|---|
| | 2021 | 2022 | % Change Actual | 2021 | 2022 | % Change Actual Cost | % Change – CDD | % Change – Food Covers |
| **January** | 43,292 | 44,658 | +3.2% | 336,700 | 347,240 | +3.1% | +15.8% | +22.3% |
| **February** | 50,837 | 45,298 | -10.9% | 395,180 | 352,330 | -10.8% | -90.8% | +31.6% |
| **March** | 55,668 | 44,914 | -19.3% | 432,400 | 349,260 | -19.2% | -6.9% | -34.3% |
| **April** | 65,511 | 51,640 | -21.2% | 509,000 | 401,500 | -21.1% | -11.4% | -37.5% |
| **May** | 77,506 | 55,628 | -28.2% | 612,200 | 432,480 | -29.4% | -32.6% | -48.1% |
| **June** | 66,115 | 45,805 | -30.7% | 513,500 | 356,090 | -30.7% | -1.0% | -55.1% |
| **July** | 70,716 | 38,031 | -46.2% | 549,160 | 295,890 | -46.1% | +4.1% | -95.5% |
| **August** | 62,738 | 49,076 | -21.8% | 487,380 | 381,480 | -21.7% | +1.7% | -34.6% |
| **September** | | | | | | | | |
| **October** | | | | | | | | |
| **November** | | | | | | | | |
| **December** | | | | | | | | |
| **Total/ Average** | **492,383** | **375,050** | **-23.8%** | **3,835,520** | **2,916,270** | **-24.0%** | **-10.3%** | **-35.4%** |

***Important note**: Cooling Degree Days (CDDs) and Food Covers are the main variables taken into consideration on Water consumption normalization.*

Energenz summarized the key items which could result in water consumption decrease below:

- Hotel Tower water consumption decreased by 41.3% and occupancy has decreased by 34.7% when compared with the same month last year.
- Water Feature consumption decreased by 24.5% when compared with the same month last year due to the changes in operations during COVID low occupancy periods.
- CT make up water consumption decreased by 16% due to the decrease in the Chiller Plant loading when compared with the same month last year.

Table 31 presents the comparison of water rates in the years 2021 and 2022. The water rate in August 2022 increased by **0.1%** compared to the same month in 2021. The average utility rate of water has decreased by **0.1%** compared to 2021.

**Table 31. Cotai - Summary of Actual Monthly Water Rates**

| Monthly Period | Year 2021 – Water Rate (MOP/m³) | Year 2022 – Water Rate (MOP/m³) | % Change – Water Rate |
|---|---|---|---|
| **January** | 7.777 | 7.776 | -0.02% |
| **February** | 7.77 | 7.78 | +0.1% |
| **March** | 7.77 | 7.78 | +0.1% |
| **April** | 7.770 | 7.775 | +0.1% |
| **May** | 7.90 | 7.77 | -1.6% |
| **June** | 7.767 | 7.774 | +0.1% |
| **July** | 7.77 | 7.78 | +0.2% |
| **August** | 7.768 | 7.773 | +0.1% |
| **September** | | | |
| **October** | | | |
| **November** | | | |
| **December** | | | |
| **Total/Average** | **7.79** | **7.78** | **-0.1%** |

Table 32 summarized the monthly normalized water consumption. The actual water consumption in August 2022 decreased by **17.5%** compared to 2019 normalized consumption in the same month.

**Table 32. Cotai - Summary of Normalized Monthly Water Consumption**

| Monthly Period | 2019 Normalized Water Consumption (m³) | 2022 Actual Water Consumption (m³) | % Change - Normalized |
|---|---|---|---|
| **January** | 48,361 | 44,658 | -7.7% |
| **February** | 49,015 | 45,298 | -7.6% |
| **March** | 49,249 | 44,914 | -8.8% |
| **April** | 51,413 | 51,640 | +0.4% |
| **May** | 54,331 | 55,628 | +2.4% |
| **June** | 56,037 | 45,805 | -18.3% |
| **July** | 54,785 | 38,031 | -30.6% |
| **August** | 59,454 | 49,076 | -17.5% |
| **September** | | | |
| **October** | | | |
| **November** | | | |
| **December** | | | |
| **Total/Average** | **422,644** | **375,050** | **-11.3%** |

## 3.5 Cotai Year 2022 Projects

Energenz will perform analysis and coordinate with MGM to implement the following projects in 2022 and reduce the site energy and water consumption which supports Energenz annual normalized energy savings target.

**Table 33. MGM Cotai Year 2022 Projects**

| Item | Initiatives |
|---|---|
| **Electricity & NG** | |
| 1 | Chiller Plant Control Optimization (CHWP/CDWP/CT) |
| 2 | Hot Water System Upgrade<br>1. Increase Max. No of Plate for Pre-heat Compact Unit<br>2. Upgrade Booster Compact Unit<br>3. Install Electric Heater for Booster Calorifier |
| 3 | AHU Static Pressure & Supply Air Temperature Set-Point Reset (Phase 2) |
| 4 | PICCV Implementation (Phase 3) |
| 5 | Hotel Towers Building Pressurization and EAF Optimization |
| 6 | Kitchen Exhaust Fan (KEF) Steps Speed Control Optimization |
| 7 | Carpark Ventilation Fans Control Optimization with IoT Sensor Installation (Phase 2) |
| 8 | VAV Space Temperature Setpoint Optimization with Heatmap Control |
| 9 | Heatmap Phasing Implementation (Stage 5, 6, 7, 8a, 8b, 10b) |
| 10 | Centralized RO Pump Set Replacement (10 Units) |
| **Water** | |
| 11 | Extension of CDP Water Recycling |
| 12 | Rainwater Harvesting for CT Make Up (Phase 1) |
| 13 | CT Conductivity Blowdown Optimization |

# Appendix A

## Regression Analysis

The 2019 normalized consumption modelling is remained used for tracking the agreed normalized energy savings target of 4% accurately for MGM Macau and MGM Cotai. The following tables show a summary of the selected independent variables for each utility which are used for the normalized consumption modelling in 2022.

### MGM Macau

| Utilities | 2022 | |
|---|---|---|
| **Electricity (kWh)** | CDD (Base T=17°C) | |
| **LPG (kg)** | Min OAT | |
| **Water ($m^3$)** | CDD (Base T=17°C) | Food Cover |

### MGM Cotai

| Utilities | 2022 | |
|---|---|---|
| **Electricity (kWh)** | CDD (Base T=17°C) | |
| **Natural Gas ($m^3$)** | Min OAT | |
| **Water ($m^3$)** | CDD (Base T=17°C) | Food Cover |

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